Mid-Submission: Confidence-Based Query Routing for Hybrid Retrieval

1. Project

We test whether cheap per-query features can route each query to BM25, dense, or tuned hybrid retrieval, and escalate low-confidence queries to a cross-encoder under a small budget. Evaluation uses held-out 5-fold predictions and paired bootstrap CIs on NFCorpus, SciFact, FiQA, CQADupStack-Unix and Quora (3,343 queries). The metric is nDCG@10.

2. Hypotheses

Phase 1 (original):

	Hypothesis	Verdict	Evidence
H1	Routing beats hybrid and RRF	Not supported (vs hybrid)	Router −0.0073 [−0.0100, −0.0045]; with τ escalation −0.0054; cost-aware −0.0012 [−0.0025, +0.0002]. Beats RRF by +0.0142, but so does hybrid.
H2	Recall@100 matches hybrid	Supported	All routers within ±0.01.
H3	Budgeted escalation keeps most of the rerank gain	Not supported	At a 10% budget it keeps 16% of the +0.0067 blended gain.
H4	Latency near hybrid	Partly	Router +5–13 ms. 10% escalation is 2.5–7× hybrid. Always-rerank costs 368–406 ms.

Phase 2 (tentative and subject to change, added to try to explain the Phase 1 hypotheses failing).
Natural queries show no routing signal, so we ask when routing can work.

- **P2-H1:** routing needs query groups whose best retriever differs; natural sets have none.
- **P2-H2:** the 18 features separate keyword from paraphrase queries, so route labels are learnable.
- **P2-H3:** one shared α is a poor compromise on a mixed-style workload.
- **P2-H4:** router gain is lowest at a balanced style mix (due to majority of queries getting routed to fixed hybrid) and higher at either extreme.

3. What we have built

- BM25, dense (e5-small-v2), hybrid and RRF retrievers
- 18 query and retrieval-signal features
- GBM, logistic, advantage-regression and cost-aware routers
- Cross-encoder escalation with predicted-gain, confidence and τ-threshold rules
- Evaluation tooling: oracle headroom, measured latency, risk-coverage analysis, ablations

4. Results so far

Phase 1: always-hybrid (0.5880) is the best dependable cheap policy. The GBM router (0.5808) is not better than random routing at the same proportions (+0.0010 [−0.0016, +0.0036]).

5. What's Next

- Our next step is to run the pipelines on a different kind of dataset and check if it outperforms fixed hybrid on that, and if the results validate our Phase 2 hypotheses.